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SDS-Mamba: Structure-aware Dynamic Scanning Mamba for Hyperspectral and Multispectral Image Fusion

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***Abstract*—Hyperspectral-multispectral image fusion (HMIF) aims to reconstruct high-fidelity high spatial-resolution hyperspectral images (HR-HSI) by integrating the spatial information of high-resolution multispectral images (HR-MSI) and the spectral information of low-resolution hyperspectral images (LR-HSI). Recently, Mamba-based methods have been increasingly introduced into HMIF tasks due to their linear complexity and long-range modeling capability. However, existing methods mostly rely on fixed scanning strategies, which may force spatially adjacent but structurally inconsistent heterogeneous regions to be sequentially modeled, while weakening effective dependency modeling between spatially distant yet structurally similar regions. Consequently, spatial detail degradation and spectral distortion occur. To address this issue, we propose a Structure-aware Dynamic Scanning Mamba (SDS-Mamba) network. The core idea of SDS-Mamba is to dynamically reconstruct structure-aware scanning sequences to model and adaptively fuse global consistency features and local detail features for high-fidelity reconstruction. Specifically, we first introduce a Global Structure-aware Mamba Modeling (GSMM) Module, which constructs a global scanning sequence based on structural similarity to capture reliable global spatial–spectral consistency features. Then, a Local Boundary-aware Mamba Modeling (LBMM) Module is designed to build boundary-aware continuous local scanning sequences, effectively preserving local edge and detail features. Finally, a Global–Local Discrepancy-aware Fusion (GLDF) Module leverages discrepancy features to guide the effective integration of global and local features. Extensive experiments on multiple benchmark datasets demonstrate that SDS-Mamba achieves SOTA performance in both quantitative and visual comparisons.**

***Index Terms*—Hyperspectral and Multispectral Image Fusion, Deep Learning, Mamba, State Space Models (SSMs).**

I. Introduction

HIGH spatial-resolution hyperspectral images (HR-HSI) can simultaneously provide rich continuous spectral information and fine spatial structural details, and have important application value [1]–[3] in remote sensing tasks such as object recognition [4], fine classification [5], and change detection [6]. However, due to the inherent constraints of sensor imaging mechanisms, it is difficult to directly acquire observation data with both high spatial resolution and high spectral resolution. Therefore, Hyperspectral–multispectral image fusion (HMIF) has been proposed, aiming to combine the high spectral fidelity of low-resolution hyperspectral images (LR-HSI) and the spatial detail representation ability of high-resolution multispectral images (HR-MSI), so as to reconstruct high-fidelity HR-HSI [7]–[11].

Early traditional HMIF methods mainly rely on handcrafted prior models such as component substitution (CS) and multi-resolution analysis (MRA). These methods are usually based on linear imaging assumptions or fixed degradation models. Although they have strong interpretability, they have difficulty characterizing the nonlinear spatial–spectral coupling relationships in complex land-cover scenes [12]–[14]. With the development of deep learning, convolutional neural networks (CNNs) and Transformers have gradually become mainstream methods [15]–[18]. CNNs can effectively extract spatial textures and edge details through local receptive fields, but their ability to characterize large-scale land-cover structures and long-range dependencies is limited. Transformers rely on the self-attention mechanism to establish global dependencies, which alleviates the locality limitation of CNNs to a certain extent. However, their computational complexity grows quadratically with the number of tokens, resulting in large computational and memory costs when modeling high-resolution remote sensing images. Therefore, how to effectively model long-range spatial-spectral dependencies while maintaining computational efficiency remains a key issue in HMIF.

In recent years, state space models (SSMs), especially Mamba models, have shown great potential in visual tasks due to their linear complexity and powerful long-range sequence modeling capability [19], [20]. Unlike traditional SSMs, Mamba dynamically adjusts the state update and information retention process according to the input content through the selective state space mechanism, thereby achieving efficient long-distance dependency modeling with low computational cost. Subsequently, many studies have attempted to extend Mamba to two-dimensional visual scenarios. The core idea is to map two-dimensional image features into one-dimensional token sequences through specific scanning strategies, so as to adapt to the sequence modeling paradigm of Mamba [21], [22].

However, although existing visual Mamba methods have improved the sequential modeling of two-dimensional visual features through multi-directional scanning, local-window scanning, and continuous scanning strategies, as illustrated in Fig. 1(a)–(d), these methods mainly design token traversal

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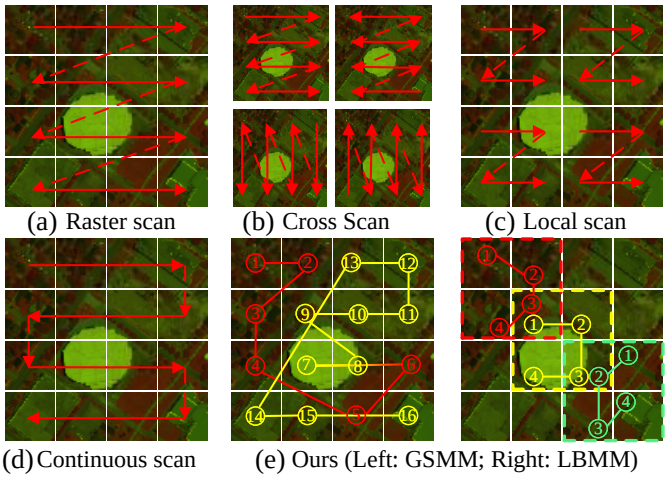


Fig. 1. Comparison of different state propagation strategies for HMIF. (a) Raster scan. (b) Cross scan. (c) Local scan. (d) Continuous scan. (e) The proposed scanning strategy. Existing scanning strategies usually rely on fixed or predefined traversal orders and lack adaptive awareness of image structures. Raster scan may force hidden states to propagate across structurally inconsistent regions, introducing irrelevant spatial–spectral information and causing structural degradation. Although cross scan, local scan, and continuous scan can expand propagation directions or preserve local continuity to some extent, they still struggle to simultaneously model spatially distant but structurally similar regions and local boundary details. In contrast, we construct a structure-similarity-aware global scanning sequence to model spatially distant but structurally similar regions (Left), and a boundary-aware local continuous scanning sequence to preserve edge and texture details (Right). Compared with conventional scanning strategies, our method can simultaneously model global structural consistency and local detail fidelity, thereby facilitating high-fidelity HR-HSI reconstruction.

from the perspectives of scanning direction, spatial range, or spatial continuity. Their scanning sequences are still primarily determined by predefined spatial relationships, lacking a mechanism to adaptively organize tokens according to the spatial–spectral structures of the input features. In HMIF, such structure-agnostic state propagation may, on the one hand, sequentially model spatially adjacent regions with significantly different structural characteristics, thereby impairing the preservation of local boundaries and texture details. On the other hand, it is difficult to establish effective dependencies between spatially distant yet structurally similar regions, limiting the modeling of global spatial–spectral consistency and consequently leading to blurred boundaries, degraded textures, and spectral distortions. Therefore, how to overcome the constraints imposed by predefined spatial relationships on scanning sequences and exploit input structural information to adaptively construct state propagation paths, while simultaneously maintaining global structural consistency and preserving local structural details, is the key issue investigated in this work.

Based on the above analysis, we propose Structure-aware Dynamic Scanning Mamba (SDS-Mamba). The core idea of SDS-Mamba is to introduce an input-dependent structure-aware scanning strategy. Unlike predefined spatial traversal orders, the proposed strategy generates scanning sequences according to the structural relationships of the current input features, enabling hidden states to propagate along paths that match the input structures and thereby achieving structure-aware state propagation. As illustrated in Fig. 1(e), the proposed strategy is applied to both global (Left) and local (Right) modeling processes. Specifically, the Global Structure-aware Mamba Modeling (GSMM) Module organizes spatially distant but structurally similar regions into adjacent sequence positions, facilitating state propagation among structurally consistent regions to capture reliable global spatial–spectral features. The Local Boundary-aware Mamba Modeling (LBMM) Module constructs boundary-aware scanning sequences within overlapping local windows, enabling state propagation to better focus on object boundaries, texture variations, and fine-grained local structures, thereby preserving local structural details. Finally, the Global–Local Discrepancy-aware Fusion (GLDF) Module dynamically guides feature fusion according to the response discrepancies between global and local features, adaptively balancing global consistency modeling and local detail preservation.

Our contributions are as follows:

- We analyze the limitations of existing visual Mamba methods for HMIF from the perspective of state propagation paths. Their predefined scanning sequences lack explicit awareness of input structures, which may lead to sequential modeling of spatially adjacent but structurally inconsistent regions while hindering effective interactions between spatially distant but structurally similar regions, thereby degrading fusion performance.
- We propose a structure-aware dynamic scanning strategy that generates scanning sequences according to the structures of the current input features, allowing different inputs to produce different state propagation paths and thereby alleviating structural degradation caused by predefined scanning paths.
- Based on the proposed strategy, we develop SDS-Mamba. GSMM and LBMM introduce structure-aware dynamic scanning into global and local modeling, respectively, to capture complementary global and local features. GLDF further adaptively integrates these two types of features to achieve high-fidelity HR-HSI reconstruction.
- Experimental results demonstrate that SDS-Mamba achieves state-of-the-art performance on four public datasets.

II. RELATED WORK

A. Traditional methods

Traditional HMIF methods mainly rely on manually designed priors and explicit optimization models [23]–[25]. According to their modeling strategies, they can be roughly divided into two categories. The first category is spatial-detail injection-based methods, including CS and MRA, which introduce spatial details from HR-MSI into LR-HSI through component replacement or multiscale decomposition to enhance spatial resolution [26]. For example, [27] employs wavelet transform to inject the spatial information of HR-MSI into the multiscale representation of LR-HSI, achieving high spatial–spectral resolution reconstruction. The second category formulates HMIF by establishing observation degradation models and incorporating low-dimensional priors, such as sparse representation, matrix factorization, and tensor decomposition, for

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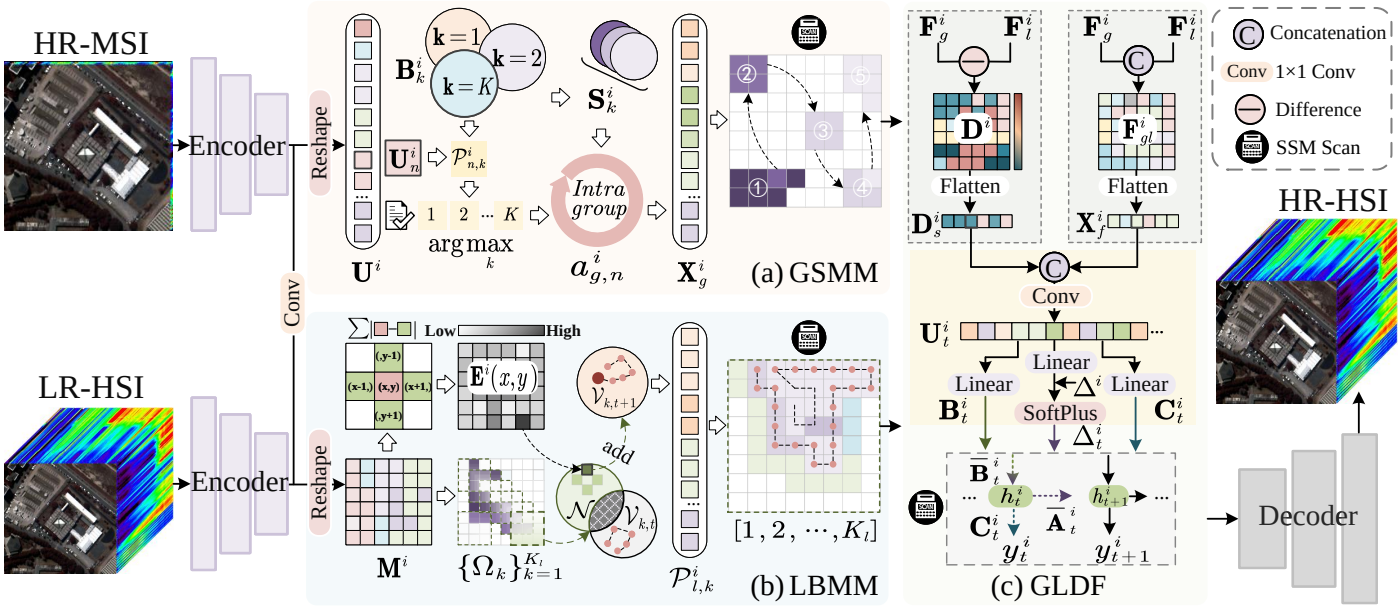


Fig. 2. Overview of the proposed SDS-Mamba network. Given HR-MSI and LR-HSI as inputs, the network first extracts multi-scale spatial-spectral features through two modality-specific encoders. The fused features are then fed into three key modules: (a) the Global Structure-aware Mamba Modeling (GSMM) module, which constructs a global structure-aware scanning sequence according to token-level structural similarity, to obtain reliable global features; (b) the Local Boundary-aware Mamba Modeling (LBMM) module, which builds boundary-aware continuous local scanning paths within overlapping windows to preserve local edges and textures; and (c) the Global-Local Discrepancy-aware Fusion (GLDF) module, which injects the discrepancy between global and local features into the Mamba parameter generation process for adaptive feature fusion. Finally, the decoder progressively aggregates the fused features to reconstruct the high-fidelity HR-HSI.

optimization-based reconstruction [28]–[30]. For instance, [31] models HR-HSI as a three-dimensional tensor and utilizes coupled sparse tensor decomposition to jointly represent LR-HSI and HR-MSI, thereby preserving spatial–spectral correlations. Recently, [32] proposed a discrete–continuous MAP framework for semi-blind fusion that implicitly accommodates unknown spatial degradation through a data-dependent prior and mitigates residual spectral mismatch through spectral gain correction, without explicitly estimating a blur kernel. Although traditional methods offer strong physical interpretability, some remain constrained by predefined degradation models and handcrafted priors, limiting their ability to capture nonlinear spatial–spectral coupling in complex scenes and potentially resulting in blurred edges, loss of texture details, or spectral distortion.

B. Deep learning methods

In recent years, with the development of deep learning, CNNs and Transformers have gradually become mainstream methods in HMIF tasks. Compared with traditional methods, deep learning methods can directly learn the nonlinear mapping from LR-HSI and HR-MSI to HR-HSI from data, thereby significantly improving reconstruction accuracy. Early CNN-based methods usually adopt dual-branch or multi-branch structures to separately extract and fuse spectral and spatial features [33]–[35]. For example, [36] constructs spatial and spectral reconstruction branches using CNNs to compensate for the missing spatial information of LR-HSI and the missing spectral information of HR-MSI. However, the receptive field of CNNs is usually local and fixed, which limits their ability to model long-range dependencies. Subsequently, Transformers were introduced into HMIF tasks to model long-range spatial structures and global spectral correlations [37]–[39]. For example, [40] employs a recurrent Transformer to model cross-modal spatial–spectral interactions; [41] introduces spatial–spectral edge enhancement and partitioned Transformer mechanisms to alleviate edge errors and enhance global–local spatial modeling capability; and [42] models spatial–spectral dependencies through a local–global collaborative Transformer. Although Transformers have strong global modeling capability, the complexity of self-attention grows quadratically with the number of tokens, which brings considerable computational overhead in HMIF tasks with high spectral dimensionality. Therefore, how to reduce computational complexity while maintaining long-range dependency modeling capability remains a key issue for the further development of HMIF.

C. State Space Models and Mamba

In recent years, SSMs have attracted increasing attention due to their ability to model long-sequence dependencies with linear complexity. Compared with the quadratic computational complexity of Transformers, SSMs can achieve efficient sequence modeling through state recurrence. Furthermore, Mamba introduces a selective state space mechanism into the SSM framework, which dynamically regulates information retention, forgetting, and propagation according to the input content, thereby enabling efficient long-range dependency modeling while maintaining $O(N)$ complexity [43]–[48]. Since two-dimensional images can be converted into one-dimensional sequences through scanning or flattening operations, Mamba has also been gradually introduced into

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visual tasks and provides a new modeling perspective for HMIF with high spectral dimensionality. For example, [49] extends Mamba to visual backbone networks and aggregates contextual information along multi-directional paths through a two-dimensional selective scanning module; [50] combines a Transformer-Mamba MoE framework, 3D Mamba scanning, and multi-scale reconstruction mechanisms to improve HSI-MSI fusion performance; [51] combines Mamba with implicit neural representation to achieve point-wise spatial–spectral fusion in the continuous domain; and [52] and [53] utilize Mamba to model long-range spatial–spectral dependencies and multi-level spatial–spectral relationships, respectively. These methods demonstrate the potential of Mamba for efficient long-range spatial–spectral modeling in HMIF tasks.

Although the above methods have demonstrated the potential of Mamba for efficient spatial–spectral modeling in HMIF, their scanning sequences are still primarily constructed according to predefined spatial relationships, lacking explicit awareness of the input structure. Different from these methods, this work focuses on structure-aware dynamic scanning and adaptively constructs state propagation paths according to the structural relationships of the input.

III. Methodology

A. Preliminaries of State Space Models and Mamba

State Space Models (SSMs) provide an efficient framework for sequence modeling by recursively updating hidden states. Given a continuous input signal $\mathbf{x}(t)$, the continuous-time SSM can be formulated as:

$$\mathbf{h}'(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}\mathbf{x}(t), \quad \mathbf{y}(t) = \mathbf{C}\mathbf{h}(t) + \mathbf{D}\mathbf{x}(t), \tag{1}$$

where $\mathbf{x}(t)$, $\mathbf{h}(t)$, and $\mathbf{y}(t)$ denote the input, hidden state, and output at time t, respectively. $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ represent the state transition matrix, input projection matrix, output projection matrix, and skip connection, respectively. To handle discrete input sequences, the continuous system is discretized with a time step Δ, yielding:

$$\mathbf{h}_t = \bar{\mathbf{A}}\mathbf{h}_{t-1} + \bar{\mathbf{B}}\mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}\mathbf{h}_t + \mathbf{D}\mathbf{x}_t, \tag{2}$$

where $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ denote the discretized state transition and input projection matrices. With zero-order hold discretization, they are defined as:

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right)\Delta\mathbf{B}. \tag{3}$$

Through recurrent state updates, SSMs can capture long-range dependencies with linear complexity. However, conventional SSMs use fixed state-space parameters during inference, which limits their input-adaptive modeling capability. To address this limitation, Mamba introduces a selective mechanism into SSMs, where several state-space parameters are dynamically generated according to the current input token. The selective state update is formulated as:

$$\mathbf{h}_t = \bar{\mathbf{A}}_t\mathbf{h}_{t-1} + \bar{\mathbf{B}}_t\mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}_t\mathbf{h}_t. \tag{4}$$

Specifically, the input-dependent parameters are generated by:

$$\begin{aligned} \Delta_t &= \mathrm{SoftPlus}\left(\Delta + \mathrm{Linear}_{\Delta}(\mathbf{x}_t)\right), \\ \mathbf{B}_t &= \mathrm{Linear}_B(\mathbf{x}_t), \quad \mathbf{C}_t = \mathrm{Linear}_C(\mathbf{x}_t), \end{aligned} \tag{5}$$

textcolorvioletwhere Δ is the learnable step-size bias. The SoftPlus function ensures a positive input-dependent step size, $\Delta_t > 0$, for discretization. Accordingly, Mamba can dynamically control information preservation, updating, and forgetting during state propagation.

Although the selective mechanism enables input-dependent state updates, visual Mamba still requires a scanning strategy to convert two-dimensional image features into one-dimensional sequences. Therefore, the scanning order directly determines the state propagation path. In HMIF tasks, a structure-agnostic scanning order may cause hidden states to propagate across heterogeneous regions and weaken effective interaction between spatially distant yet structurally similar regions. This motivates us to construct structure-aware dynamic scanning sequences from the perspective of input token organization, rather than modifying the SSM operator itself, for more adaptive spatial–spectral state propagation.

B. Overall Architecture

SDS-Mamba aims to fully exploit global–local complementary features by constructing structure-aware dynamic state propagation paths and to achieve high-fidelity HR-HSI reconstruction through adaptive feature integration. As shown in Fig. 2, SDS-Mamba mainly consists of three stages. The LR-HSI is first upsampled to the same spatial resolution as the HR-MSI using bilinear interpolation. Two 3×3 convolutional layers are then employed for feature extraction, yielding multi-scale feature representations $f_X^i = \{f_{HSI}^i, f_{MSI}^i\}$, where $i = 1, 2, 3$. Subsequently, at the i-th scale, the features from the two modalities are preliminarily fused and fed into GSMM and LBMM for selective state space modeling, respectively. GSMM constructs an input-dependent global structure-aware scanning path to model global spatial–spectral consistency, while LBMM constructs an input-dependent boundary-aware scanning path within overlapping local windows to preserve local boundaries, textures, and fine-grained structures, thereby capturing faithful local boundary and edge details. Finally, GLDF adaptively integrates global and local features by modulating state propagation parameters according to response differences, producing fused features with both global structural consistency and local detail fidelity. The fused features are progressively aggregated and enhanced through a decoder with residual refinement, ultimately reconstructing a high-fidelity HR-HSI.

C. Global Structure-aware Mamba Modeling (GSMM) Module

GSMM aims to model the global spatial–spectral dependencies among spatially distant yet structurally related regions. In HMIF, high-quality HR-HSI reconstruction requires global spatial–spectral consistency, particularly for homogeneous or similar land-cover regions distributed at different spatial locations. However, existing Mamba-based methods generally rely on predefined and input-structure-independent fixed scanning paths. Such structure-agnostic propagation orders may weaken effective information transfer between spatially distant but structurally similar regions, thereby limiting the capability

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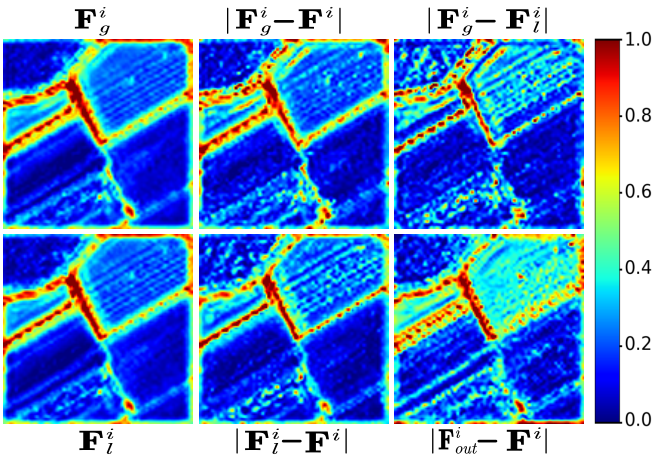


Fig. 3. Visualization of intermediate responses and response variations at the first stage. $\mathbf{F}_g^i$ and $\mathbf{F}_l^i$ denote the global structure-enhanced feature generated by GSMM and the local boundary-enhanced feature generated by LBMM, respectively. $|\mathbf{F}_g^i - \mathbf{F}^i|$ and $|\mathbf{F}_l^i - \mathbf{F}^i|$ represent the response variations introduced by GSMM and LBMM with respect to the initial fused feature $\mathbf{F}^i$. $|\mathbf{F}_g^i - \mathbf{F}_l^i|$ denotes the global–local discrepancy response, while $|\mathbf{F}_{out}^i - \mathbf{F}^i|$ shows the final feature adjustment produced by GLDF. Warmer colors indicate stronger responses. It can be observed that GSMM and LBMM introduce different feature variations. The global–local discrepancy response is more pronounced in structurally complex regions and relatively weak in homogeneous regions, providing effective guidance for balancing global spatial–spectral consistency and local detail fidelity during adaptive feature fusion.

of modeling global spatial–spectral consistency. To address this limitation, we propose the Global Structure-aware Mamba Modeling (GSMM) Module, as illustrated in Fig. 2(a). GSMM first computes token responses from the fused spatial–spectral features using fixed random projections. These responses serve as feature-space proxies for structural relationships and determine token group assignments, with tokens assigned to the same group placed consecutively in the scanning sequence. Subsequently, tokens within the same structural group are sorted according to their intra-group consistency scores to generate a global structure-aware scanning path, enabling Mamba to propagate states along the reorganized global structural sequence and obtain feature representations with global spatial–spectral consistency.

Specifically, the i-th stage features f_{HSI}^i and f_{MSI}^i are first concatenated along the channel dimension and integrated through a 3×3 convolution to obtain the fused feature $\mathbf{F}^i \in \mathbb{R}^{C_i \times H_i \times W_i}$. Subsequently, $\mathbf{F}^i$ is flattened into $\mathbf{Z}^i$ and mapped into a token sequence $\mathbf{U}^i \in \mathbb{R}^{N_i \times C_i}$, where $N_i = H_i W_i$. To arrange tokens with similar structural responses at neighboring positions in the scanning sequence, we introduce $K = 4$ fixed random projection matrices $\{\mathbf{B}_k^i\}_{k=1}^K$, each of size $C_i \times C_i$. At initialization, the entries of each matrix are independently sampled from a standard normal distribution using a local random number generator seeded with $3407 + C_i$. Each matrix is then divided by its Frobenius norm to control differences in the overall scale across matrices; for numerical stability, the normalization denominator is lower-bounded by 10^{-6}. The resulting matrices remain fixed after initialization. The response score of the n-th token under the k-th fixed random projection matrix is defined as:

$$\rho_{n,k}^i = \left\| \mathbf{U}_n^i \mathbf{B}_k^i \right\|_2^2. \tag{6}$$

Subsequently, the response scores of the n-th token under all fixed random projection matrices are normalized to obtain its response weights for different structural groups, based on which its final structural group index is determined:

$$\mathbf{\Pi}_{g,nk}^i = \frac{\exp(\rho_{n,k}^i/\eta)}{\sum_{j=1}^K \exp(\rho_{n,j}^i/\eta)}, \qquad g_n^i = \arg\max_k \mathbf{\Pi}_{g,nk}^i, \tag{7}$$

where $\mathbf{\Pi}_{g,nk}^i$ denotes the response weight of the n-th token to the k-th structural group, g_n^i denotes the index of its final assigned structural group, and $\eta = 1$ is adopted in this work.

To further determine the scanning order within the same structural group, we first compute the statistical response center of each structural group:

$$\mathbf{s}_k^i = \frac{\sum_{n=1}^{N_i} \mathbf{\Pi}_{g,nk}^i \left(\mathbf{U}_n^i \mathbf{B}_k^i \odot \mathbf{U}_n^i \mathbf{B}_k^i \right)}{\sum_{n=1}^{N_i} \mathbf{\Pi}_{g,nk}^i + \epsilon}, \tag{8}$$

where $\odot$ denotes element-wise multiplication, and ϵ is a small constant introduced to avoid numerical instability, which is set to 10^{-6} in this work. Then, the consistency score between each token and the statistical response center of its assigned structural group is calculated as:

$$a_{g,n}^i = \left\langle \mathbf{U}_n^i \mathbf{B}_{g_n^i}^i \odot \mathbf{U}_n^i \mathbf{B}_{g_n^i}^i, \mathbf{s}_{g_n^i}^i \right\rangle. \tag{9}$$

Finally, the global scanning path $\mathcal{P}_g^i$ is generated based on g^i and a_g^i as $\mathcal{P}_g^i = \mathrm{SortBy}(g^i, -a_g^i)$. Specifically, tokens are first grouped according to their structural group indices and then sorted within each group in descending order of their consistency scores. Since both the structural responses and the structural group assignments are determined by the current input features, different inputs can produce distinct token permutations, thereby forming input-dependent scanning sequences. After that, $\mathbf{Z}^i$ is reordered according to $\mathcal{P}_g^i$ to obtain the structure-aware sequence $\mathbf{X}_g^i = \mathrm{Reorder}(\mathbf{Z}^i, \mathcal{P}_g^i)$, which is fed into the SSM for global state propagation:

$$\widetilde{\mathbf{X}}_g^i = \mathrm{SSM}(\mathbf{X}_g^i). \tag{10}$$

Finally, the output sequence is restored to the original spatial layout by $\mathbf{F}_g^i = \mathrm{InverseReorder}(\widetilde{\mathbf{X}}_g^i, \mathcal{P}_g^i)$, producing the global structure-enhanced feature. As shown in Fig. 3, GSMM introduces global structure-aware adjustment to the initial fused feature and exhibits clear responses around region-level structural patterns.

D. Local Boundary-aware Mamba Modeling (LBMM) Module

LBMM aims to model local fine-grained structural details, including object boundaries and land-cover textures. In the HMIF task, high-fidelity HR-HSI reconstruction relies not only on global spatial–spectral consistency, but also on the preservation of clear object boundaries and fine-grained texture details. However, existing Mamba-based methods typically rely on predefined scanning paths that are fixed and independent of the input structure for state propagation, which can easily introduce cross-region state contamination when the scanning sequence traverses different land-cover regions, while also limiting the ability to adapt to complex local structural variations. To address this issue, we propose the

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Local Boundary-aware Mamba Modeling (LBMM) Module, as illustrated in Fig. 2(b). LBMM computes local structural difference responses between neighboring positions to identify boundary-sensitive regions and texture variations, and generates a local boundary-aware scanning path. This enables Mamba to perform effective state propagation within locally similar regions while reducing cross-region propagation at locations with significant structural differences, thereby mitigating boundary contamination and enhancing the fidelity of local structural details, such as textures and fine-grained edges.

Specifically, we first compress the fused feature $\mathbf{F}^i$ using a 1×1 convolution to obtain $\mathbf{M}^i \in \mathbb{R}^{1\times H_i\times W_i}$. Then, the structural difference response between each position and its neighboring positions is calculated as:

$$\mathbf{E}^i(x,y) = \sum_{(u,v)\in\mathcal{N}(x,y)} \left|\mathbf{M}^i(x,y) - \mathbf{M}^i(u,v)\right|, \tag{11}$$

where $\mathcal{N}(x,y)$ denotes the local neighborhood of position (x,y). A larger $\mathbf{E}^i(x,y)$ indicates a more significant local structural variation, whose response arises not only from boundaries between different land-cover regions but also from texture variations and fine-grained structural differences within the same region. To preserve the continuity of local boundaries, we divide the feature map into K_l overlapping local windows $\{\Omega_k\}_{k=1}^{K_l}$. The local window size is set to 8×8 with a sliding stride of $S = 4$, resulting in a 50% overlap between adjacent windows. For the k-th window, the position with the strongest structural difference response is selected as the scanning starting point:

$$p^i_{k,1} = \arg\max_{(x,y)\in\Omega_k} \mathbf{E}^i(x,y). \tag{12}$$

During the scanning process, the next position is selected from the unvisited neighboring positions with the largest structural difference response:

$$p^i_{k,t+1} = \arg\max_{(u,v)\in\mathcal{N}(p^i_{k,t})\cap\Omega_k\setminus\mathcal{V}_{k,t}} \mathbf{E}^i(u,v), \tag{13}$$

where $\mathcal{V}_{k,t}$ denotes the set of visited tokens in the k-th window up to step t, and $p^i_{k,t} = (x_t, y_t)$ denotes the current scanning position. When there is no unvisited token in the current neighborhood, the unvisited token with the largest structural difference response in Ω_k is reselected as a new local starting point, and the neighborhood-based scanning continues until all tokens in the window are visited. Therefore, the boundary-aware continuous scanning path of the k-th window is denoted as $\mathcal{P}^i_{l,k} = [p^i_{k,1}, p^i_{k,2}, \cdots, p^i_{k,|\Omega_k|}]$. Since $\mathbf{E}^i$ dynamically varies with the input features, the local token ordering and the resulting scanning sequence are input-dependent. Subsequently, the sequence within the k-th window, i.e., $\mathbf{F}^i_{\Omega_k}$, is reordered as $\mathbf{X}^i_{l,k}$ according to $\mathcal{P}^i_{l,k}$, and then fed into the SSM for local state propagation:

$$\widetilde{\mathbf{X}}^i_{l,k} = \mathrm{SSM}(\mathbf{X}^i_{l,k}). \tag{14}$$

Finally, the output sequences of all local windows are restored to their original window layouts according to the corresponding scanning paths. For overlapping regions, the window outputs are accumulated and averaged according to the actual number of overlaps. Through the overlap aggregation operation $\mathrm{OverAgg}(\cdot)$, the local boundary-enhanced feature $\mathbf{F}^i_l$ is obtained. As shown in Fig. 3, LBMM introduces local boundary-aware adjustment to the initial fused feature, especially around land-cover boundaries, texture transitions, and fine-grained structural variations.

E. Global-Local Discrepancy-aware Fusion (GLDF) Module

GLDF aims to effectively integrate complementary global and local features. However, simply concatenating or adding these two types of features is insufficient to fully exploit their complementarity, which may lead to unclear boundaries or spectral distortion. To address this issue, we introduce GLDF, as shown in Fig. 2(c). GLDF first computes the discrepancy feature between the global feature $\mathbf{F}^i_g$ and the local feature $\mathbf{F}^i_l$, denoted as $\mathbf{D}^i = \left|\mathbf{F}^i_g - \mathbf{F}^i_l\right|$, which characterizes the inconsistency between the two types of features. As shown in Fig. 3, the discrepancy feature is more pronounced in structurally complex regions, while remaining relatively weak in homogeneous regions. Therefore, it provides effective guidance for adaptive fusion: regions with large discrepancies require stronger local detail preservation and boundary refinement, whereas regions with small discrepancies should maintain global spatial–spectral consistency and avoid unnecessary local perturbations.

Specifically, we first concatenate $\mathbf{F}^i_g, \mathbf{F}^i_l \in \mathbb{R}^{C_i\times H_i\times W_i}$ along the channel dimension and employ a 1×1 convolution to map the channel dimension from $2C_i$ to C_i, yielding the shallow fused feature $\mathbf{F}^i_{gl} \in \mathbb{R}^{C_i\times H_i\times W_i}$. Subsequently, $\mathbf{F}^i_{gl}$ is flattened along the spatial dimension into a spatial token sequence $\mathbf{X}^i_f \in \mathbb{R}^{N_i\times C_i}$, where $N_i = H_iW_i$. Meanwhile, $\mathbf{D}^i$ is flattened in the same order to obtain $\mathbf{D}^i_s \in \mathbb{R}^{N_i\times C_i}$. For the t-th token, the discrepancy-guided input is constructed as:

$$\mathbf{U}^i_t = \mathrm{Conv}\left([\mathbf{X}^i_{f,t}, \mathbf{D}^i_{s,t}]\right), \tag{15}$$

where $\mathbf{X}^i_{f,t}$ and $\mathbf{D}^i_{s,t}$ correspond to the same spatial position, with $t = 1, \ldots, N_i$. After concatenation, the feature dimension is reduced from $2C_i$ to C_i through feature mapping, yielding $\mathbf{U}^i_t$. Different from the standard Mamba, where the selective state-space parameters are generated only from the fused token $\mathbf{X}^i_{f,t}$, GLDF injects the global–local discrepancy into the parameter generation process through $\mathbf{U}^i_t$. The parameters are generated as follows:

$$\begin{aligned}
\Delta^i_t &= \mathrm{SoftPlus}\left(\Delta^i + \mathrm{Linear}_\Delta(\mathbf{U}^i_t)\right),\\
\mathbf{B}^i_t &= \mathrm{Linear}_B(\mathbf{U}^i_t),\\
\mathbf{C}^i_t &= \mathrm{Linear}_C(\mathbf{U}^i_t).
\end{aligned} \tag{16}$$

where Δ^i denotes the learnable step-size bias at the i-th stage. Subsequently, the fused token sequence $\mathbf{X}^i_f$ is fed into the SSM for state propagation with the state-space parameters generated from $\mathbf{U}^i_t$, enabling interactive fusion between global and local features:

$$\widetilde{\mathbf{X}}^i_f = \mathrm{SSM}\left(\mathbf{X}^i_f; \Delta^i_t, \mathbf{B}^i_t, \mathbf{C}^i_t\right). \tag{17}$$

Then, the output sequence is restored to the original spatial layout to obtain the fused feature $\mathbf{F}^i_{out}$. As shown in Fig. 3,

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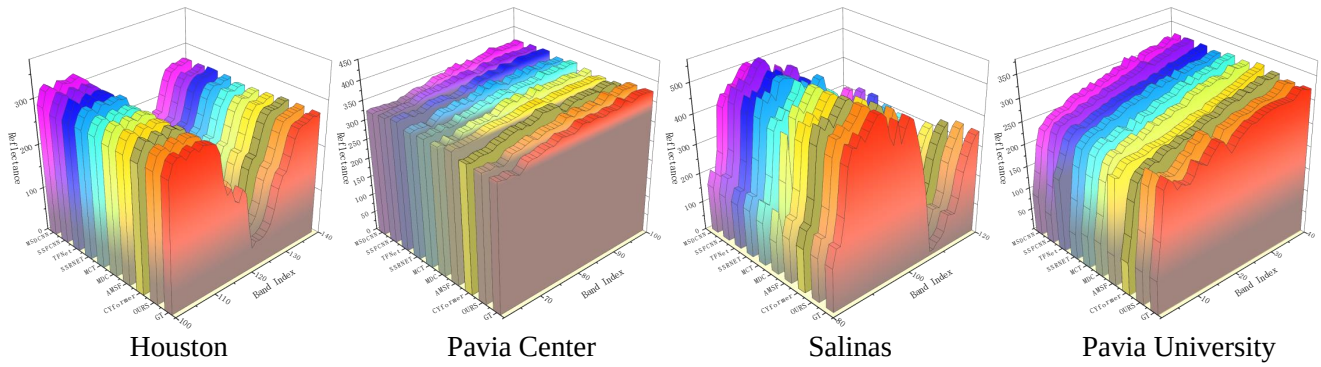


Fig. 4. Spectral reflectance curves within a 40-band interval at the spatial location $(70, 70)$ across four datasets for different models. Compared with other methods, Ours produces spectral curves that align most closely with the GT, demonstrating superior spectral fidelity.

GLDF introduces discrepancy-aware adaptive recalibration to the initial fused feature, especially in boundary regions, mixed pixels, and structurally complex regions, thereby effectively balancing global structural consistency and local detail fidelity.

F. Loss Function

Let $\mathbf{Y} \in \mathbb{R}^{H\times W\times C}$ and $\hat{\mathbf{Y}} \in \mathbb{R}^{H\times W\times C}$ denote the ground-truth HR-HSI and the reconstructed fused image, respectively. To train the proposed network, we adopt the simple and widely used mean squared error (MSE) loss as the optimization objective. Specifically, no additional spectral, spatial, perceptual, or adversarial constraints are introduced. This simple objective directly measures the pixel-wise reconstruction discrepancy across all spectral bands, thereby ensuring stable convergence during training. The MSE loss is defined as:

$$\mathcal{L}_{MSE} = \frac{1}{HWC}\|\hat{\mathbf{Y}} - \mathbf{Y}\|_2^2. \tag{18}$$

IV. Experiment

A. Datasets

To comprehensively evaluate the fusion reconstruction capability of the proposed method under different remote sensing scenarios, experiments are conducted on four public hyperspectral remote sensing datasets, including Houston, Salinas, Pavia Center, and Pavia University. These datasets differ in imaging sensors, spatial resolutions, numbers of spectral bands, and land-cover types, covering typical remote sensing scenarios such as urban areas, agricultural scenes, and campus environments. Therefore, they can provide a comprehensive evaluation of the generalization capability of the model in terms of spatial detail recovery and spectral information preservation.

1) Houston Dataset: The Houston dataset was acquired by the ITRES CASI-1500 hyperspectral imager, covering the University of Houston and its surrounding urban area in Texas, USA. This dataset contains 144 spectral bands with a spectral range from 364 nm to 1046 nm. The image size is 349×1905, and the spatial resolution is 2.5 m. Since this scene contains various land-cover types, such as buildings, roads, grass, and trees, with complex spatial structures, it is suitable for evaluating the spatial structure recovery and spectral preservation capabilities of fusion models in urban scenes.

2) Salinas Dataset: The Salinas dataset was collected by the AVIRIS sensor over the Salinas Valley agricultural area in California, USA. The original data contain 224 spectral bands. After removing the bands seriously affected by water absorption and noise, 204 effective bands are commonly retained for experiments. The spatial size of this dataset is 512×217, with a spatial resolution of 3.7 m. Since it contains multiple crop categories with subtle spectral differences among different land-cover types, this dataset can be used to evaluate the spectral preservation ability of the model and its discrimination capability for similar land-cover classes.

3) Pavia Center Dataset: The Pavia Center dataset was acquired by the ROSIS sensor over the city center of Pavia, Italy. After preprocessing, this dataset contains 102 effective spectral bands. The image size is 1096×715, and the spatial resolution is 1.3 m. This scene includes dense buildings, roads, vegetation, and other land-cover types, with rich spatial textures and boundary structures. Therefore, it is suitable for evaluating the detail recovery capability of the model in high-spatial-resolution urban scenes.

4) Pavia University Dataset: The Pavia University dataset was also acquired by the ROSIS sensor, covering the University of Pavia and its surrounding areas in Italy. This dataset contains 103 effective spectral bands, with a spatial size of 610×340 and a spatial resolution of 1.3 m. Compared with Pavia Center, this scene has more diverse land-cover distributions, including roads, buildings, grass, trees, and bare soil. Therefore, it is commonly used to verify the spatial reconstruction capability and spectral consistency preservation ability of models in complex campus scenes.

B. Experimental Setup

1) Experimental Settings and Implementation Details: To ensure consistency in experimental evaluation, the same data partition and degradation generation strategy are adopted for all datasets. Specifically, for each hyperspectral dataset, a central image patch with a size of 128×128 is selected as the testing region, while the remaining regions are used for model training. During training, a high-spatial-resolution hyperspectral image patch with a size of 128×128 is randomly

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TABLE I
QUANTITATIVE COMPARISON ON HOUSTON AND SALINAS DATASETS. [KEY: BEST, SECOND BEST]

Methods	Houston					Salinas				
	RMSE↓	PSNR↑	ERGAS↓	SAM↓	SSIM↑	RMSE↓	PSNR↑	ERGAS↓	SAM↓	SSIM↑
MSDCNN [33]	1.5644	44.0543	2.5461	2.8766	0.9841	2.3791	39.9441	2.9526	1.7880	0.7994
SSFCNN [34]	1.2180	46.0258	3.3736	2.2205	0.9744	1.8795	41.9913	8.8923	1.7214	0.8017
TFNet [35]	1.1594	46.4540	1.6503	1.9874	0.9848	1.7978	42.3744	2.2862	1.3943	0.8372
SSRNET [36]	1.2059	46.1120	1.7256	2.2158	0.9813	2.1483	40.8306	5.4396	1.8227	0.7713
MCT [37]	1.0592	47.2392	1.4434	1.8686	0.9867	2.0372	41.2941	5.1777	1.5975	0.8161
MDC [38]	0.8175	49.4889	1.1854	1.4131	0.9903	1.7029	42.8483	3.6782	1.4363	0.8286
AMSF [39]	1.2386	45.8799	1.6469	2.0859	0.9844	2.3617	40.0076	5.5984	1.7134	0.7896
CYformer [40]	0.8080	49.5908	1.2231	1.4657	0.9913	1.4162	44.4496	2.5594	1.2325	0.8531
MCIFNet [51]	0.8377	49.2771	1.2099	1.3944	0.9909	1.3433	44.909	2.0948	1.1123	0.9097
DHMF-Net [53]	0.7885	49.803	1.1156	1.2304	0.9917	1.1649	46.1468	1.2541	0.8335	0.9012
Ours	0.7082	50.7355	1.0243	1.2937	0.9923	1.0444	47.0940	1.1107	0.7493	0.9208

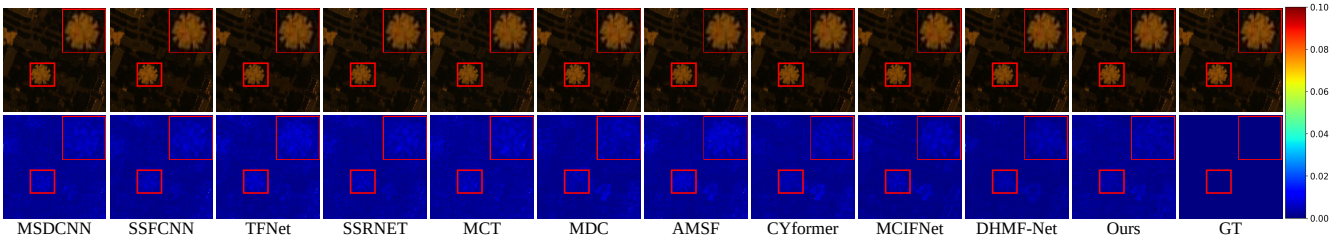


Fig. 5. Fusion results of different methods on the Houston dataset. The first row shows the reconstructed pseudo-color RGB images, where the red boxes indicate selected local regions and the enlarged views are shown in the upper-right corner. The second row presents the corresponding absolute difference maps between the reconstructed results and the ground truth.

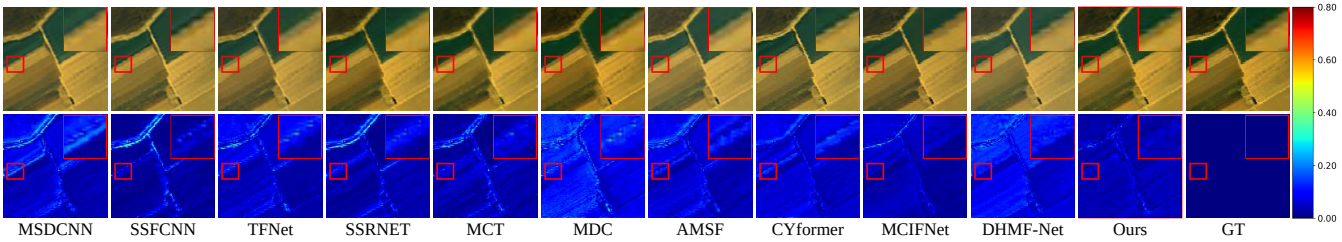


Fig. 6. Fusion results of different methods on the Salinas dataset. The first row shows the reconstructed pseudo-color RGB images, where the red boxes indicate selected local regions and the enlarged views are shown in the upper-right corner. The second row presents the corresponding absolute difference maps between the reconstructed results and the ground truth.

cropped from the training region at each iteration and used as the ground truth (GT). Then, the GT is degraded to generate the model inputs. For LR-HSI, the GT is first spatially blurred using a 5×5 Gaussian kernel and then downsampled with a scaling factor of 4. For HR-MSI, 5 representative bands are uniformly selected from the GT through spectral degradation. Finally, the generated LR-HSI and HR-MSI are used as the model inputs, while the GT serves as the supervision signal for model optimization.

All experiments are conducted on a computing platform equipped with an NVIDIA GeForce RTX 4090 GPU. The deep learning framework is PyTorch 2.0.0, accelerated by CUDA 12.7. SDS-Mamba is trained using the Adam optimizer with an initial learning rate of 1×10^{-4}. For fair comparison, all competing methods are run under the same hardware and software environments, and their default training configurations are followed as much as possible.

2) Evaluation Metrics: To comprehensively evaluate the spatial detail recovery and spectral information preservation capabilities of the fused images, five commonly used quantitative metrics are adopted in this work, including RMSE, PSNR, SSIM, ERGAS, and SAM. Specifically, RMSE measures the average reconstruction error between the reconstructed image and the reference image, while PSNR reflects the signal-to-noise ratio of the reconstructed image. SSIM evaluates the structural preservation ability from the perspectives of luminance, contrast, and structural information. ERGAS is used to measure the overall reconstruction error of multi-band images, reflecting the global fusion quality. SAM evaluates the degree of spectral distortion by calculating the angular difference between spectral vectors, and is an important metric for assessing spectral fidelity. These metrics provide a comprehensive evaluation of different methods from the perspectives of pixel-wise error, structural consistency, global quality, and

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TABLE II
QUANTITATIVE COMPARISON ON PAVIA CENTER AND PAVIA UNIVERSITY DATASETS.

Methods	Pavia Center					Pavia University				
	RMSE↓	PSNR↑	ERGAS↓	SAM↓	SSIM↑	RMSE↓	PSNR↑	ERGAS↓	SAM↓	SSIM↑
MSDCNN [33]	3.4609	37.3469	3.9300	4.2764	0.9832	2.2135	41.3345	1.6068	2.5280	0.9519
SSFCNN [34]	3.2149	37.9841	3.7132	3.7411	0.9854	2.4321	40.7012	2.3431	2.8160	0.9412
TFNet [35]	3.2627	37.8594	3.7112	4.2800	0.9817	1.8218	42.6057	1.3528	2.1426	0.9757
SSRNET [36]	3.2661	37.8503	3.7254	3.7257	0.9850	1.6278	43.5835	1.2090	1.9473	0.9807
MCT [37]	2.8099	39.1571	3.1503	3.7569	0.9860	1.5696	43.9001	1.1736	1.8874	0.9815
MDC [38]	2.9177	38.8478	3.3424	3.7399	0.9859	1.5581	43.9639	1.1737	1.8665	0.9816
AMSF [39]	3.4098	37.4761	3.6987	5.0685	0.9713	1.9789	41.6901	1.4806	2.4026	0.9747
CYformer [40]	2.6994	39.5056	3.0357	3.5937	0.9863	1.5605	43.9507	1.1729	1.8700	0.9821
MCIFNet [51]	3.0662	38.3988	3.3358	3.3831	0.9869	1.6061	43.7001	1.1735	1.8292	0.9803
DHMF-Net [53]	3.1204	38.2467	3.6249	3.1943	0.9875	1.4477	44.2282	1.1334	1.8386	0.9838
Ours	2.6076	39.8061	2.9942	3.3795	0.9877	1.4331	44.6904	1.0914	1.7277	0.9828

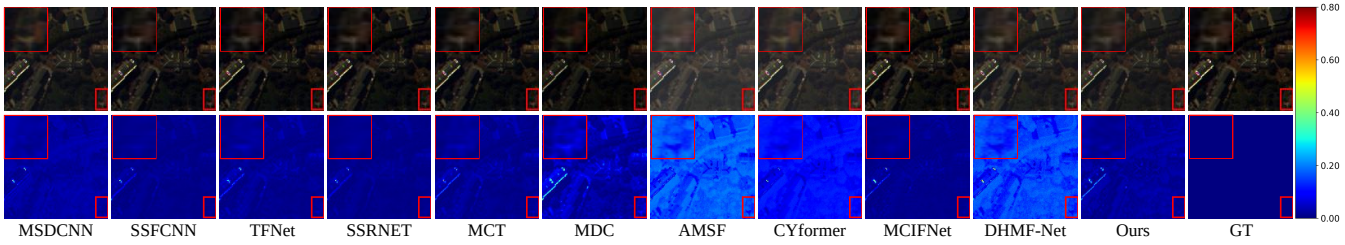


Fig. 7. Fusion results of different methods on the Pavia Center dataset. The first row shows the reconstructed pseudo-color RGB images, where the red boxes indicate selected local regions and the enlarged views are shown in the upper-right corner. The second row presents the corresponding absolute difference maps between the reconstructed results and the ground truth.

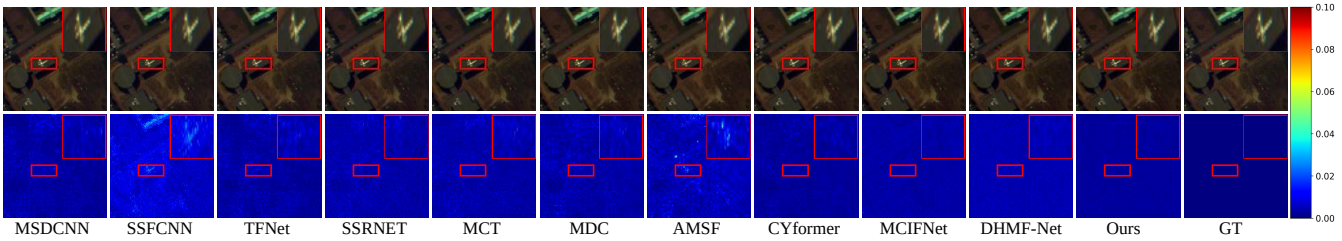


Fig. 8. Fusion results of different methods on the Pavia University dataset. The first row shows the reconstructed pseudo-color RGB images, where the red boxes indicate selected local regions and the enlarged views are shown in the upper-right corner. The second row presents the corresponding absolute difference maps between the reconstructed results and the ground truth.

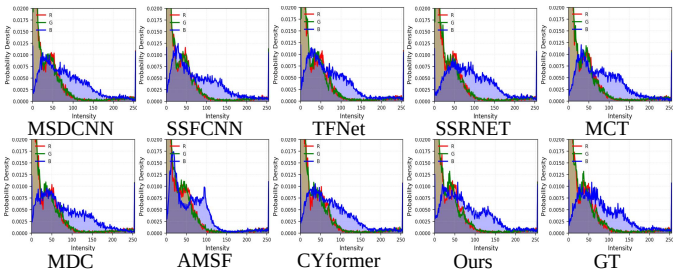


Fig. 9. The comparison of pseudo-RGB intensity probability density distributions on the Houston dataset is presented. The R, G, and B curves represent the probability density distributions of three selected representative spectral bands in each reconstructed result, respectively. Compared with other methods, SDS-Mamba exhibits intensity distributions that are overall closer to those of GT, indicating that the proposed method can better preserve the radiometric consistency and spectral distribution characteristics of the reconstructed image.

spectral fidelity.

C. Comparison with SOTA Methods

1) Quantitative Comparisons: As shown in Table I, SDS-Mamba achieves highly competitive performance on the Houston and Salinas datasets. On the Houston dataset, SDS-Mamba obtains the best RMSE, PSNR, ERGAS, and SSIM values of 0.7082, 50.7355 dB, 1.0243, and 0.9923, respectively, while achieving the second-best SAM of 1.2937. Compared with the second-best results, SDS-Mamba reduces RMSE by 0.0803 and ERGAS by 0.0913, while improving PSNR by 0.9325 dB. On the Salinas dataset, SDS-Mamba achieves the best performance across all five metrics, with an RMSE of 1.0444, PSNR of 47.0940 dB, ERGAS of 1.1107, SAM of 0.7493, and SSIM of 0.9208. Compared with the second-best results, SDS-Mamba reduces RMSE by 0.1205 and ERGAS by 0.1434, while improving PSNR by 0.9472 dB. Unlike fixed scanning strategies that rely on predefined spatial orders, SDS-Mamba dynamically organizes state propagation sequences

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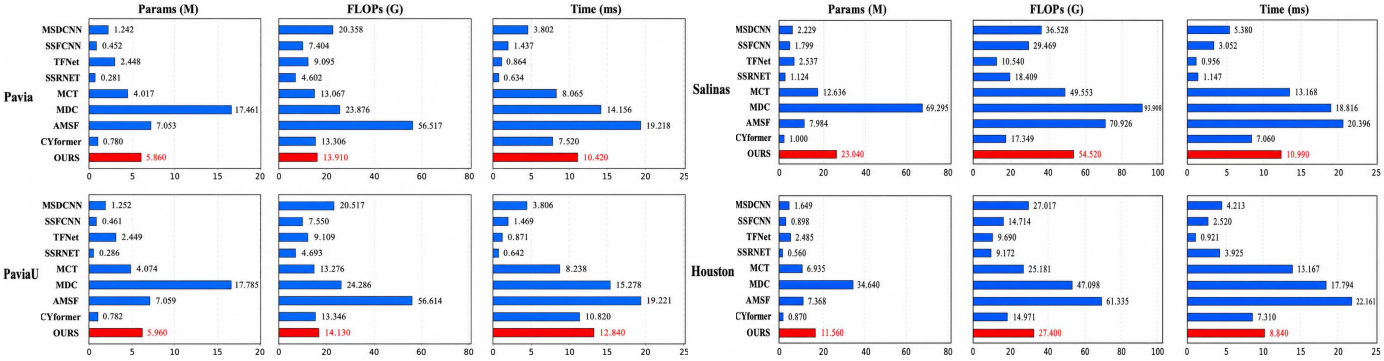


Fig. 10. Complexity comparison of different methods on the Pavia Center, Pavia University, Salinas, and Houston datasets in terms of parameters, FLOPs, and inference time. Red bars indicate the proposed SDS-Mamba. The results show that SDS-Mamba achieves competitive computational efficiency while maintaining superior reconstruction accuracy.

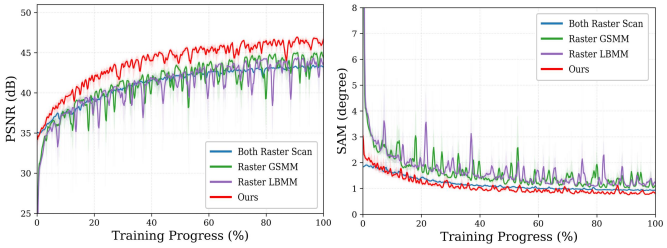


Fig. 11. Effectiveness of the proposed structure-aware dynamic scanning strategy on the Salinas dataset. Both Raster Scan denotes the variant that replaces both GSMM and LBMM with conventional raster scan; Raster GSMM and Raster LBMM replace the scanning strategy of GSMM or LBMM with raster scan, respectively; Ours denotes the proposed structure-aware dynamic scanning strategy. The proposed method obtains higher PSNR and lower SAM, verifying its effectiveness in spatial-spectral dependency modeling.

according to the input structures. GSMM maintains global spatial–spectral consistency by enhancing state propagation among structurally similar regions, while LBMM preserves local structural details by reducing feature mixing across heterogeneous boundaries. GLDF further adaptively integrates the two complementary representations. Consequently, SDS-Mamba effectively alleviates spectral distortion and spatial detail loss, as evidenced by the lower SAM and ERGAS values and the reconstructed spectral curves closer to the GT in Fig. 4.

Table II further demonstrates the effectiveness of SDS-Mamba on the Pavia Center and Pavia University datasets. On the Pavia Center dataset, SDS-Mamba achieves the best RMSE, PSNR, ERGAS, and SSIM values of 2.6076, 39.8061 dB, 2.9942, and 0.9877, respectively, while obtaining the second-best SAM of 3.3795. Compared with the second-best results, SDS-Mamba reduces RMSE by 0.0918 and ERGAS by 0.0415, while improving PSNR by 0.3005 dB. On the Pavia University dataset, SDS-Mamba achieves the best RMSE, PSNR, ERGAS, and SAM values of 1.4331, 44.6904 dB, 1.0914, and 1.7277, respectively, together with the second-best SSIM of 0.9828. Compared with the second-best results, SDS-Mamba reduces RMSE by 0.0146, ERGAS by 0.0420, and SAM by 0.1015, while improving PSNR by 0.4622 dB. Pavia Center and Pavia University contain complex land-cover structures and dense building boundaries, where fixed scanning strategies may lead to state propagation across structurally inconsistent regions. By dynamically organizing token adjacency, SDS-Mamba enhances state propagation among structurally similar regions and along locally continuous structures, thereby reducing spectral response mixing and local detail degradation. As shown in Fig. 4, the reconstructed spectra are more consistent with the GT, which is also supported by the lower SAM and ERGAS values.

2) Qualitative Comparisons: As shown in Figs. 5–8, SDS-Mamba achieves more faithful visual reconstruction results on the Houston, Salinas, Pavia Center, and Pavia University datasets. In the reconstructed pseudo-color RGB images, several competing methods exhibit blurred boundaries, color deviations, or incomplete structural details, particularly in the highlighted local regions. In contrast, SDS-Mamba preserves clearer spatial structures and more consistent local textures, producing results that are visually closer to the GT. This advantage can be further observed in the corresponding absolute difference maps. Compared with other methods, SDS-Mamba generally produces weaker and more spatially concentrated error responses, indicating its effectiveness in suppressing reconstruction errors around complex boundaries and heterogeneous regions. These visual improvements are consistent with the structure-aware dynamic scanning mechanism of SDS-Mamba. Unlike fixed scanning strategies that may propagate states across heterogeneous regions, LBMM constructs locally continuous paths along boundaries and textures, thereby reducing cross-boundary response mixing and spatial detail loss. Meanwhile, GSMM organizes structurally similar tokens into adjacent sequence positions, facilitating state propagation among consistent regions and alleviating spatial–spectral deviations. Consequently, SDS-Mamba produces lower error responses around complex structures while preserving both local details and global spatial–spectral consistency.

In addition to the visual reconstruction results, Fig. 9 presents a comparison of the pseudo-RGB intensity probability distributions on the Houston dataset. The distribution produced by SDS-Mamba is closer to that of the GT than those of the competing methods, indicating better preservation of the radiometric distribution and spectral response characteristics. Several competing methods exhibit noticeable distribution shifts or unstable fluctuations, which may correspond to color distortions or spectral inconsistencies. In contrast,

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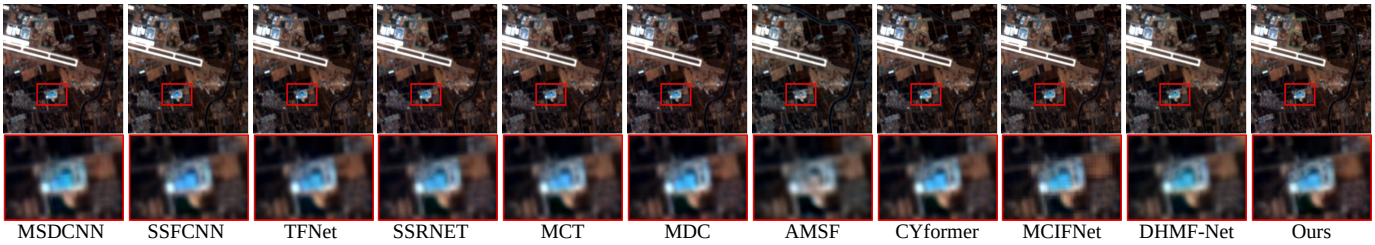


Fig. 12. Qualitative comparison of different HMIF methods on the real GF5/GF6 HSI–MSI dataset. The red boxes indicate the representative regions selected for enlarged visualization.

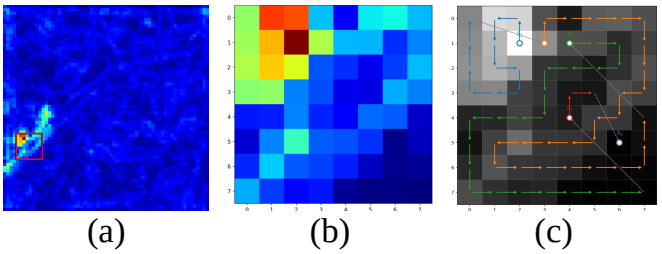


Fig. 13. Visualization of the LBMM scanning process on the Pavia Center dataset. (a) Boundary response map, where the black box marks the representative local window selected for visualization. (b) Boundary responses within the selected (8×8) window. (c) The corresponding LBMM scanning path. Colored solid arrows indicate locally continuous 4-neighbor scanning segments, while gray dashed lines denote path restarts after the current segment terminates. The results show that LBMM primarily propagates states along spatially continuous local structures while using limited restarts to complete the traversal of the remaining high-response tokens.

TABLE III
MODULE ABLATION EXPERIMENT RESULTS OF SDS-MAMBA ON PAVIA CENTER DATASET.

GSMM	LBMM	GLDF	RMSE	PSNR	ERGAS	SAM	SSIM
	✓	✓	2.9996	38.5896	3.3858	3.8499	0.9857
✓		✓	3.1850	38.0686	3.5936	3.8904	0.9848
✓	✓		3.4108	37.6561	3.8072	4.2071	0.9822
✓	✓	✓	**2.6076**	**39.8061**	**2.9942**	**3.3795**	**0.9877**

SDS-Mamba maintains a distribution more consistent with the GT across different intensity ranges. This distribution-preserving capability further supports the above mechanism analysis: by reducing state propagation across structurally inconsistent regions while strengthening effective dependencies among structurally similar regions, the structure-aware dynamic scanning strategy suppresses the mixing and propagation of irrelevant responses during state recurrence, resulting in more stable spatial–spectral responses. This observation is consistent with the lower error responses shown in Figs. 5–8, further demonstrating that SDS-Mamba improves spatial detail reconstruction while maintaining high spectral fidelity.

3) Computational Efficiency Comparisons: To evaluate the computational efficiency of the proposed method, we compare the number of parameters, FLOPs, and inference time of different HMIF models on four datasets, as shown in Fig. 10, where the red bars denote SDS-Mamba. Overall, SDS-Mamba achieves a favorable balance between reconstruction accuracy and computational cost. Compared with complex fusion models such as MDC and AMSF, SDS-Mamba requires fewer parameters and lower FLOPs on most datasets while maintaining superior reconstruction performance. This advantage mainly benefits from its compact architecture and efficient Mamba-based spatial–spectral modeling mechanism. Specifically, Transformer-based methods usually rely on global self-attention to model long-range dependencies, which often introduces high computational complexity. In contrast, Mamba captures long-range spatial–spectral dependencies through efficient state propagation, thereby reducing the additional overhead caused by dense attention computation. Although some lightweight CNN-based methods require fewer parameters or FLOPs, their limited ability to model global spatial–spectral correlations leads to relatively inferior reconstruction accuracy. Therefore, SDS-Mamba achieves a better trade-off between accuracy and efficiency, demonstrating its potential for large-scale hyperspectral–multispectral image fusion.

D. Ablation Study

To evaluate the effectiveness of each key component in SDS-Mamba, we conduct module ablation experiments on the Pavia Center dataset, as shown in Table III. The investigated modules include GSMM, LBMM, and GLDF. When GSMM or LBMM is removed, the corresponding enhanced feature is replaced with the initial fused feature; when GLDF is removed, the discrepancy-aware fusion is replaced with simple concatenation-based fusion to keep the overall framework consistent. The experimental results show that removing any module leads to performance degradation. Without GSMM, RMSE and SAM increase, while PSNR and SSIM decrease, indicating that global structure-aware modeling helps capture long-range spatial–spectral dependencies. Removing LBMM also degrades the model performance, demonstrating the importance of local boundary-aware scanning for preserving fine spatial details and edge structures. When GLDF is replaced with simple concatenation-based fusion, all metrics deteriorate noticeably, suggesting that discrepancy-aware fusion can more effectively coordinate global consistency and local detail preservation. In contrast, the complete SDS-Mamba achieves the best results across all metrics, verifying the complementarity of GSMM, LBMM, and GLDF and their effectiveness for high-quality HR-HSI reconstruction.

1) Effectiveness of Structure-aware Dynamic Scanning: To validate the effectiveness of the proposed structure-aware dynamic scanning strategy, SDS-Mamba is compared with three Raster Scan-based variants on the Salinas dataset, as shown in Fig. 11. From the PSNR curves, Both Raster

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TABLE IV
ABLATION EXPERIMENT RESULTS OF GLDF ON PAVIA CENTER DATASET.

	RMSE	PSNR	ERGAS	SAM	SSIM
w/o $\mathbf{D}^i$	2.8936	38.9021	3.2527	3.7421	0.9860
w/ GLDF	**2.6076**	**39.8061**	**2.9942**	**3.3795**	**0.9877**

Scan converges more slowly and achieves the lowest final reconstruction accuracy, indicating that fixed raster scanning is insufficient to model the complex spatial–spectral dependencies in HMIF tasks. In contrast, Ours maintains higher PSNR throughout training, demonstrating that the structure-aware dynamic token arrangement provides Mamba with a more reasonable state propagation path and thus improves spatial reconstruction accuracy. The SAM curves further show that Ours consistently obtains lower spectral angular errors, suggesting that the proposed dynamic scanning strategy also enhances spectral fidelity. A further comparison between Raster GSMM and Raster LBMM shows that replacing GSMM with Raster Scan weakens the ability to associate spatially distant but structurally similar regions, thereby degrading global spatial–spectral consistency modeling. When LBMM is replaced with Raster Scan, the model fails to continuously model local boundaries and texture variations, leading to reduced local detail preservation. Overall, Ours achieves the best convergence performance in terms of both PSNR and SAM, verifying that GSMM and LBMM improve reconstruction performance from the perspectives of global structural consistency and local boundary detail modeling, respectively.

2) Effectiveness of GLDF: To further evaluate the effectiveness of the proposed GLDF, we conduct ablation experiments on the Pavia Center dataset, as shown in Table IV. Specifically, we compare the full GLDF model with its variant without the discrepancy-guided term $\mathbf{D}^i$. The experimental results show that removing $\mathbf{D}^i$ leads to consistent performance degradation across all evaluation metrics. This indicates that simply fusing global and local features without explicit discrepancy modeling is insufficient to fully exploit their complementary information. In contrast, introducing $\mathbf{D}^i$ yields clear performance improvements across all metrics, with particularly significant gains in PSNR and SSIM. This demonstrates that the discrepancy feature effectively guides adaptive interaction between global structural consistency and local detail preservation, thereby reducing feature misalignment and improving reconstruction quality.

3) Experiment on Real-Life HSI–MSI Fusion: To further evaluate the performance of SDS-Mamba under realistic sensor conditions, we conduct experiments on a real GF5/GF6 HSI–MSI dataset. The LR-HSI is acquired by the Advanced Hyperspectral Imager (AHSI) onboard the GF-5 satellite, containing 330 spectral bands over the wavelength range of 400–2500 nm. The HR-MSI is acquired by the Wide Field of View (WFV) sensor onboard the GF-6 satellite, containing four effective bands over the wavelength range of 450–900 nm.

Since no corresponding GT is available for the real data, we evaluate the fusion results through qualitative comparisons and two no-reference quality metrics, BRISQUE [54] and NSSQS [55], where lower values indicate better performance. As shown in Fig. 12, compared with other methods, SDS-Mamba reconstructs clearer building contours and preserves more complete structural details. Moreover, as reported in Table V, SDS-Mamba achieves the best BRISQUE and NSSQS values of 74.7071 and 30.1419, respectively. These qualitative and quantitative results demonstrate that SDS-Mamba effectively preserves spatial structures and spatial–spectral consistency under real sensor observations, further validating its applicability to real-world HSI–MSI fusion.

TABLE V
QUANTITATIVE NO-REFERENCE EVALUATION OF GF5/GF6 HSI–MSI FUSION RESULTS.

Methods	BRISQUE [54] ↓	NSSQS [55] ↓
MSDCNN [33]	74.7186	33.9853
SSFCNN [34]	78.2977	35.6414
TFNet [35]	75.6562	36.4576
SSRNET [36]	76.3914	36.2482
MCT [37]	77.4328	35.1668
MDC [38]	76.3985	34.2543
AMSF [39]	76.2286	35.2109
CYformer [40]	74.9637	34.4237
MCIFNet [51]	75.5354	34.7723
DHMF-Net [53]	76.3422	33.8990
Ours	**74.7071**	**30.1419**

TABLE VI
ABLATION STUDY ON THE NUMBER OF STRUCTURAL GROUPS K IN GSMM ON THE PAVIA CENTER DATASET.

	RMSE	PSNR	ERGAS	SAM	SSIM
$K = 2$	3.0780	38.9718	3.0092	3.4290	0.9860
$K = 3$	2.8257	39.1083	3.1852	3.6075	0.9868
$K = 4$	**2.6076**	**39.8061**	**2.9942**	**3.3795**	**0.9877**
$K = 5$	2.6776	39.5433	3.0980	3.4698	0.9854
$K = 6$	3.3339	37.6718	3.7732	3.7912	0.9850

4) Analysis of LBMM Scanning Behavior: To further analyze the scanning behavior of LBMM, we visualize a representative example on the Pavia Center dataset in Fig. 13. Fig. 13(a) shows the boundary response map, while Fig. 13(b) presents the responses within a selected 8×8 local window. Based on these responses, LBMM constructs the scanning path shown in Fig. 13(c), where colored solid arrows denote locally continuous scanning segments under 4-neighbor connectivity, and gray dashed lines indicate path restarts when the current segment cannot be extended. Most tokens are traversed through locally connected paths, while each restart remains within the same window and selects an unvisited token with a high boundary response. This enables LBMM to preserve local scanning continuity while effectively covering structurally responsive regions.

5) Effectiveness of the Number of Structural Groups: To investigate the influence of the number of structural groups K in GSMM, we conduct experiments on the Pavia Center

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TABLE VII
ABLATION STUDY ON THE LOCAL WINDOW SIZE W IN LBMM ON THE PAVIA CENTER DATASET.

	RMSE	PSNR	ERGAS	SAM	SSIM
$W = 4$	2.9273	38.8014	3.3085	3.6665	0.9866
$W = 8$	**2.6076**	**39.8061**	**2.9942**	**3.3795**	**0.9877**
$W = 16$	2.7317	39.4020	3.0879	3.4231	0.9874

dataset by varying K from 2 to 6, as shown in Table VI. The results show that $K = 4$ achieves the best performance across all five metrics. A small K may group tokens with different structural characteristics together, limiting structurally consistent state propagation, whereas an excessively large K may over-fragment structurally similar tokens and weaken their effective interactions. Therefore, $K = 4$ provides an appropriate balance between structural discrimination and intra-group state propagation and is adopted as the default setting.

6) Effectiveness of the Local Window Size: To investigate the influence of the local window size W in LBMM, we conduct experiments on the Pavia Center dataset with $W = 4$, 8, and 16, as shown in Table VII. The results show that $W = 8$ achieves the best performance across all five metrics. A small W limits the spatial range for modeling continuous boundaries and local structural dependencies, whereas an excessively large W may introduce more heterogeneous structures and weaken the locality of state propagation. Therefore, $W = 8$ provides an appropriate balance between local structural coverage and spatial locality and is adopted as the default setting.

V. CONCLUSION

To overcome the limitations of existing Mamba-based methods that rely on fixed scanning paths in HMIF tasks, this paper proposes a structure-aware dynamic scanning Mamba network, termed SDS-Mamba. Specifically, GSMM and LBMM are designed to construct global structure-aware scanning paths and local boundary-aware scanning paths, respectively, enabling state propagation to follow sequence directions that better conform to image structures. In this way, SDS-Mamba effectively enhances global spatial–spectral consistency while preserving local structural details. Furthermore, GLDF is employed to adaptively fuse complementary global and local feature representations, achieving a balance among global consistency, local detail preservation, and spectral fidelity. Extensive experimental results on four benchmark datasets demonstrate that SDS-Mamba outperforms existing state-of-the-art methods in terms of both quantitative performance and visual reconstruction quality, fully validating the effectiveness of the proposed structure-aware dynamic scanning strategy for HMIF tasks.

Despite these results, SDS-Mamba relies on predefined rules for input-dependent scan-path construction, whose discrete operations limit direct gradient-based optimization of scanning decisions. Future work will explore learnable scanning strategies and differentiable approximations. Moreover, the main quantitative experiments use a fixed degradation protocol and do not establish robustness to unknown or spatially varying degradations, which requires further evaluation.

VI. ACKNOWLEDGMENT

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